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Revisiting Seismicity Criticality: A New Framework for Bias Correction of Statistical Seismology Model Calibrations

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Key Points:

- Identification and quantification of boundary effects, finite-size effects, and censorship that bias the calibrations of the Epidemic-Type Aftershock Sequences (ETAS) models
- Development of a framework for correcting biases in the branching ratio n of the ETAS model from the estimated dependence of the apparent branching ratio $n_{app}(M_{co})$ as a function of changing cut-off magnitudes M_{co}
- After corrections to obtain n_{true}, seismicity is still found subcritical ($n_{true} < n_c = 1$), however with values tending to be larger than recent reports that do not consider the above biases

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract

The Epidemic-Type Aftershock Sequences (ETAS) model and its variants effectively capture the space-time clustering of seismicity, setting the standard for earthquake forecasting. Accurate unbiased ETAS calibration is thus crucial. But we identify three sources of bias, (a) boundary effects, (b) finite-size effects, and (c) censorship, which are often overlooked or misinterpreted, causing errors in seismic analysis and predictions. By employing an ETAS model variant with variable spatial background rates, we propose a method to correct for these biases, focusing on the branching ratio n, a key indicator of earthquake triggering potential. Our approach quantifies the variation in the apparent branching ratio (n_{app}) with increased cut-off magnitude (M_{co}) above the optimal cut-off (M_{co}^{best}), which is considered the best threshold for balancing catalog completeness and the amount of available data. The $n_{app}(M_{co})$ function yields insights superior to traditional point estimates. We validate our method using synthetic earthquake catalogs, accurately recovering the true branching ratio (n_{true}) after correcting biases with $n_{app}(M_{co})$. Additionally, our method introduces a refined estimation of the minimum triggering magnitude (m_0), a crucial parameter in the ETAS model. Applying our framework to the earthquake catalogs of California, New Zealand, the China Seismic Experimental Site in Sichuan and Yunnan provinces, and Noto Peninsula in Japan, we find that seismicity hovers away from the critical point, $n_c = 1$, remaining distinctly subcritical, however with values tending to be larger than recent reports that do not consider the above biases. Understanding seismicity's critical state significantly enhances our comprehension of seismic patterns, aftershock predictability, and informs earthquake risk mitigation and management strategies.

Plain Language Summary

Seismicity can be likened to epidemic processes, where background earthquakes, primarily driven by plate tectonic forces, can initiate sequences of triggered earthquakes. These triggered events, in turn, can generate further sequences, resulting in a cascading chain of seismicity. The branching ratio n—quantifying the expected number of triggered aftershocks per event—is a key parameter for quantifying seismic clustering, forecasting cascading risks, and optimizing hazard mitigation strategies. However, biases from spatiotemporal catalog boundaries, finite samples, and magnitude cutoffs systematically skew its estimation, undermining the reliability of statistical seismology calibrations. To address this issue, we combined extensive simulation experiments with state-of-the-art cluster characterization models for parameter inversion and developed a method to correct these biases in the estimation of the branching ratio n. Applying our correction framework to earthquake catalogs from California, New Zealand, the China Seismic Experimental Site, and the Noto Peninsula in Japan confirms that seismicity is subcritical. However, the corrected branching ratio values tend to be larger than those reported in recent studies that do not account for these biases. This refined understanding of seismicity enhances our ability to characterize earthquake sequences, improve earthquake predictability, and guide more effective seismic hazard mitigation strategies.

1. Introduction

Seismicity and fault rupturing in the Earth's crust can be likened to epidemic processes, as exemplified by the Epidemic-Type Aftershock Sequences (ETAS) model (Kagan, 1991; Kagan & Knopoff, 1981, 1987; Musmeci & Vere-Jones, 1992; Ogata, 1988, 1998; Ogata & Zhuang, 2006), wherein background earthquakes ("immigrants"), assumed to be driven by the forces of plate tectonics, have the potential to trigger cohorts of earthquakes, referred to as first-generation "daughters." These first-generation events, possessing fertility, can in turn act as

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"mothers," triggering second-generation "daughters" and creating a cascading effect (Felzer et al., 2002, 2003; Hawkes & Oakes, 1974; Helmstetter & Sornette, 2002; Zhuang et al., 2002, 2005).

In this dynamic process, a crucial parameter known as the branching ratio n plays a pivotal role in reflecting the state of the crustal system. This parameter signifies both the mean number of fertile first-generation "daughters" triggered per "mother" and the fraction of all triggered events relative to the total number of earthquakes. The transition point occurs at $n = n_c = 1$, demarcating the subcritical regime ($n < 1$), where the sequence of events is stationary, from the supercritical regime ($n > 1$), where the number of earthquakes increases exponentially over time with a finite probability (Helmstetter & Sornette, 2002). Consequently, the branching ratio n serves as an indicator of the proportion of observed earthquakes triggered by preceding events when the magnitude distribution can be separated from the other components (Zhuang et al., 2013). The extent to which n falls below the critical point n_c has significant implications on our understanding of seismicity patterns, on the forecastability of aftershocks and for earthquake hazard mitigation and risk management. More profoundly, since the introduction (Bak et al., 1987) and application of the concept of self-organized criticality to seismicity (Bak & Tang, 1989; Sornette & Sornette, 1989), a flurry of studies have proposed that earthquake processes might operate at or near a critical point (de Arcangelis et al., 2016; Fisher et al., 1997; Keilis-Borok & Soloviev, 2002; Main et al., 2006; Shcherbakov et al., 2006; Wanliss et al., 2017; Zöller et al., 2001), in the sense of critical phenomena in Statistical Physics. In contrast, research by Wu (1998) and subsequent findings in statistical seismology suggest that seismicity might be a self-organized process that does not reach a critical state, with evidence showing deviations from the critical point at $n_c = 1$ (Chu et al., 2011; Nandan, Ram, et al., 2021; Nandan et al., 2022). On the other hand, Huang et al. (1998) introduced a generalized sandpile model on a hierarchical system of faults, suggesting the possible coexistence between self-organized criticality and genuine criticality of individual large earthquakes. Whether or not they are close to such criticality is frequently seen as having profound implications for our understanding of the physics behind earthquakes. Hence, obtaining an unbiased estimation of the branching ratio n of seismicity holds significant importance.

A few previous authors have acknowledged the existence of several sources of biases in the estimation of n, which are non-stationarity and non-uniformity of the background rate when not accounted for properly, spatial and temporal boundary effects, finite-size effects and censorship. In the present study, our primary objective is to quantify the biases in the calibration of the branching ratio using the ETAS model. We develop a method to correct the apparent branching ratio n_{app} obtained from calibrations, in order to derive a good approximation of the true branching ratio n_{true}, via a procedure validated through synthetic tests. Our methodology leverages the optimal estimation of the cut-off magnitude (M_{co}^{best}) for a given catalog, which is considered the best threshold for balancing catalog completeness and the amount of available data. We focus on how n_{app} shifts when the used cut-off magnitude M_{co} is artificially increased above M_{co}^{best}, enabling us to methodically investigate the impacts of boundary effects, finite-size effects of seismicity sample and censorship. Applying this framework to earthquake catalogs from California, New Zealand, the China Seismic Experimental Site (CSES), and Noto Peninsula of Japan, we re-examine the question of whether seismicity operates near a critical point ($n_c = 1$). This reassessment carries significant implications for our understanding of seismic patterns, improving aftershock predictability, and shaping strategies for earthquake risk mitigation and hazard management.

The organization of the manuscript starts with Section 2, which presents the ETAS model, defines the different estimation methods of its branching ratio n and explains the meaning of subcriticality, criticality and supercriticality. Section 2 also describes the different effects that can bias the estimation of n and reviews existing methods for correcting these biases. Section 3 presents synthetic tests demonstrating the biases and presents our proposed correction method, which is illustrated and tested on synthetic catalogs. Section 4 describes the empirical catalogs used in this study. Section 5 examines in detail the application of our correction method to the empirical data of California, New Zealand, the CSES in Sichuan and Yunnan provinces, and Noto Peninsula of Japan. Section 6 discusses the results and concludes.

2. The ETAS Model, Calibration, and Biases

2.1. The ETAS Model

The ETAS model is an instantiation adapted to seismicity of the self-excited Hawkes stochastic point process, which is non-Markovian when the temporal memory kernel is non-exponential (Hawkes & Oakes, 1974; Kagan & Knopoff, 1981, 1987; Ogata, 1988, 1998). It enjoys two equivalent interpretations:

1. A simple ensemble branching model (Hawkes & Oakes, 1974; Kagan, 1991; Sornette & Werner, 2005b). In this conceptualization, a given sequence of events is mapped on many trees, each tree corresponding to a specification of what are the background events and what are the "mothers" and triggered "daughters." Each tree starts with a "immigrant" (or background earthquake) and all the branches that emanate from this "immigrants" correspond to the aftershocks and then aftershocks of aftershocks and so on. Each tree representation is endowed with a probability representing the likelihood of this specific association between earthquakes, given the observed sequence. The tree probabilities derive from the expression of the intensity of the ETAS model given by Formula 1 below (Zhuang et al., 2002). In this representation, the ETAS model comprises statistically independent Poisson seismicity clusters (each cluster being a tree), while there is dependence within every cluster (tree branches corresponding to all generations). Consequently, the branching ratio n is an average, encompassing not only magnitudes but also across an ensemble of realizations of the stationary point process (for $n \leq 1$) sustained by the presence of background events.
2. A coupled earthquake interaction model (Helmstetter & Sornette, 2002). In this conceptualization, each non-background event is collectively triggered by all preceding events, with each event contributing a weight determined by the fertility law (or productivity law) $F(m)$ that decays in space and time governed by the space kernel $S(x, y, m)$ and time kernel $T(t)$. The product of $F(m)$, $S(x, y, m)$, and $T(t)$ forms the triggering function. Consequently, at any given spatiotemporal point (x, y, t), the ETAS model is defined by the seismicity rate (or intensity) λ, which encapsulates the dependence on the historical seismicity H_t recorded until time t, as:

$$\lambda(t,x,y|H_t) = \mu(x,y) + \sum_{i:t_i<t} F(m_i)\, T(t-t_i)\, S\big(x-x_i, y-y_i,\, m_i\big) \tag{1}$$

The seismicity rate, at time t and position (x, y), is thus the sum of contributions from both the background intensity function, $\mu(x, y)$, and a sum over all preceding earthquakes, in which each past earthquake contributes with a multiplicative separable spatio-temporal kernel and with an amplitude given by its fertility $F(m)$. In this formulation, the branching ratio n can be interpreted as quantifying the contribution of a past earthquake to a future earthquake, averaged over an ensemble of realizations and over all magnitudes. This perspective is the only one possible for nonlinear generalizations of the Hawkes model whose triggering functions depend nonlinearly on previous events (Kanazawa & Sornette, 2021, 2023; Ouillon & Sornette, 2005; Sornette & Ouillon, 2005).

The rigorous mathematical demonstration of the equivalence of these two representations was first presented in Hawkes and Oakes (1974). Sornette & Werner (2005b) provided an intuitive understanding of the underlying mechanism for this equivalence, based on the formulation of the seismicity rate of the ETAS model as a linear sum over past earthquakes combined with the exponential form of the hazard rate of point process.

Formula 1 for the seismicity rate is complemented by the Gutenberg-Richter distribution given by

$$P(m) = \frac{b\ln(10)10^{-bm}}{10^{-bm_0} - 10^{-bm_{\max}}},\ m_0 \leq m \leq m_{\max} \tag{2}$$

where b is typically close to one. In this formula, $m_{\max}$ is the largest magnitude of earthquakes capable of triggering other earthquakes and thus the Gutenberg-Richter distribution is truncated at $m_{\max}$. It is important to stress that the ETAS model also requires an additional characteristic magnitude m_0, which is the smallest magnitude of earthquakes capable of triggering other earthquakes. In other words, the fertility of an earthquake is zero if its magnitude is smaller than m_0. It plays the role of an "ultra-violet" cut-off ensuring the convergence and stationarity of the model (Sornette & Werner, 2005a). The fertility law defining the number of direct aftershocks ("daughters") of a "mother" event of magnitude m is given by:

$$F(m) = K\, e^{\alpha\,(m-m_0)},\ m_0 \leq m \leq m_{\max} \tag{3}$$

where K and α are constants.

The other terms in the sum in the right-hand-side (r.h.s.) of Formula 1 read

$$T(t-t_i)\,S\big(x-x_i, y-y_i,\ m_i\big) = \frac{T_{\text{norm}}\cdot e^{-\frac{t-t_i}{\tau}}}{(t-t_i+c)^{1+\omega}}\cdot\frac{S_{\text{norm}}}{\Big[(x-x_i)^2+\big(y-y_i\big)^2+de^{\gamma(m_i-M_{\text{co}})}\Big]^{1+\rho}} \tag{4}$$

which are respectively the Omori-Utsu law $T(t-t_i)$ defined with the constants c, ω, and τ, and the spatial Green function with constants d, γ, and ρ, which describe the temporal and spatial dependencies within the model. T_{norm} and S_{norm} represent normalization constants for the time and space kernels, ensuring they are proper probability density functions (PDFs). This represents a common choice of spatial and temporal kernels, though other formulations are also often used (Felzer et al., 2002, 2003; Helmstetter & Sornette, 2002; Zhuang et al., 2002, 2005).

In this study, the spatial variability of background rates $\mu(x, y)$ is treated as a non-parametric function. Its estimation is through a weighted kernel function used in (Nandan, Ram, et al., 2021; Nandan et al., 2022),

$$\mu(x,y) = \frac{1}{T}\sum_{i=1}^{N} IP_i\cdot\frac{QD^{2Q}}{\pi}\,\frac{1}{\Big[(x-x_i)^2+\big(y-y_i\big)^2+D^2\Big]^{1+Q}} \tag{5}$$

which is a sum over all earthquakes weighted by their probabilities IP_i to be background events (Zhuang et al., 2002). The normalization by the duration T of the primary catalog ensures that $\mu(x, y)$ represents the seismicity rate per unit time, while the factor $\pi^{-1}QD^{2Q}$ ensures normalization per unit area at location (x, y). The choice of the power-law kernel is motivated by prior research indicating its superior performance compared to previously more commonly used Gaussian kernels (e.g., Helmstetter et al., 2007; Nandan, Kamer, et al., 2021). The estimation of the model, including model parameters $\{Q, D, K, \alpha, c, \omega, \tau, d, \gamma, \rho\}$ and nonparametric spatial variability of the background rate, $\mu(x, y)$, is accomplished through the extended expectation-maximization (EM) algorithm (Veen & Schoenberg, 2008). We refer to Nandan, Ram, et al. (2021) and Nandan et al. (2022) for details.

This ETAS model has been implemented on earthquake catalogs, including those recorded globally, in California and in New Zealand (Nandan, Ram, et al., 2021; Nandan et al., 2022). Nandan, Ram, et al. (2021) and Nandan et al. (2022) have conducted comparative assessments against other ETAS model variants through pseudo-prospective forecasting experiments to assess its superior performance in earthquake forecasting, which further advances the ETAS model as a benchmark for evaluating alternative earthquake forecasting models (Kamer et al., 2021; Nandan, Kamer, et al., 2021; Ogata, 2017).

2.2. Definition and Estimation Methods of the Branching Ratio *n*

Three methods suggested in prior literature can be used to estimate n.

2.2.1. The Formulaic Approach

The branching ratio is, by definition, the average number of fertile "daughter" per "mother." Given the fertility law presented by Formula 3, the branching ratio is the average of $F(m)$ over all possible values of $m \geq m_0$, weighted by the Gutenberg-Richter (GR) law given by Formula 2. The formula for the branching ratio reads (Sornette & Werner, 2005b)

$$n \equiv \int_{m_0}^{m_{\max}} P(m)F(m)\mathrm{d}m = \begin{cases} \dfrac{Kb\big[1-10^{-(b-a)(m_{\max}-m_0)}\big]}{(b-a)\big[1-10^{-b(m_{\max}-m_0)}\big]} & a\neq b \\[2ex] \dfrac{Kb\,\ln(10)(m_{\max}-m_0)}{1-10^{-b(m_{\max}-m_0)}} & a=b \end{cases} \tag{6}$$

The determination of the branching ratio n thus necessitates knowledge of the parameters b, m_0, $m_{\max}$, K, and $a = \alpha/\ln(10)$. In practical scenarios involving catalogs with a total of N earthquakes with magnitudes $\{m_1, m_2, \ldots, m_N\}$, Seif et al. (2017) suggested substituting the $P(m)$ in Formula 2 with the empirical frequency-magnitude distribution, leading to the following estimator

$$n = \frac{1}{N}\sum_{i=1}^{N} K\, e^{\alpha\,(m_i - m_0)}\, H(m_i - m_0) \tag{7}$$

where $H[.]$ is the Heaviside function. Consequently, this approach simplifies the estimation of the branching ratio n to rely solely on the parameters N, m_0, K, and a.

2.2.2. The Counting Approach

Helmstetter and Sornette (2003), Zhuang et al. (2013), and Seif et al. (2017) demonstrated that the branching ratio n is equivalently represented as the ratio of triggered events to the total number of earthquakes in the ETAS model when $n < 1$. This relationship is expressed by the formula

$$n = \frac{N_{\text{tri}}}{N_{\text{tri}} + N_{\text{bkg}}} \tag{8}$$

where N_{tri} and N_{bkg} denote the number of triggered and background events of magnitude larger than or equal to m_0, respectively. The calculation of n requires knowledge of either N_{tri} or N_{bkg} alone, along with the condition of knowing the total number of earthquakes N of magnitude larger than or equal to m_0 ($N = N_{\text{tri}} + N_{\text{bkg}}$). Additionally, Seif et al. (2017) proposed a method for estimating the branching ratio n by considering the proportion of aftershocks that are second or higher generations. This approach is represented as

$$n = \frac{N_{\text{tri}}^{\text{2nd+higher}}}{N_{\text{tri}}} \tag{9}$$

where $N_{\text{tri}}^{\text{2nd+higher}}$ is the number of aftershocks of magnitude larger than or equal to m_0 that are of generation larger than or equal to 2. In other words, $N_{\text{tri}}^{\text{2nd+higher}}$ is the number of aftershocks that are themselves triggered by previous aftershocks. This approach requires knowledge of the branching structure of seismicity.

2.2.3. The Mean-Variance-Based Estimation Approach

Hardiman and Bouchaud (2014) introduced a parameter-independent method for approximating the branching ratio n. This technique relies exclusively on the mean ($\bar{N}$) and variance (σ^2) of the event count within a suitably large time window, as indicated by their sample estimates:

$$\bar{N} = \frac{1}{W}\sum_{i=1}^{W} N(i) \tag{10}$$

$$\sigma^2 = \frac{1}{W-1}\sum_{i=1}^{W}\left[N(i) - \bar{N}\right]^2 \tag{11}$$

where $N(i)$ represents the event count in the ith time window out of a total of W. The branching ratio can be approximated as (Hardiman & Bouchaud, 2014)

$$n \approx 1 - \sqrt{\frac{\bar{N}}{\sigma^2}} \tag{12}$$

This method has been tested using simulated data and empirical financial market data. It is found to be fragile in the presence of a non-constant and/or non-uniform distribution of background events.

2.3. Subcriticality, Criticality, or Supercriticality of Seismicity

In statistical seismology, the branching ratio n is primarily studied using the formulaic and counting approaches. Seif et al. (2017) calibrated the ETAS model with a spatially varying background rate, yielding a branching ratio close to or slightly larger than 1 in Southern California and Italy by using Formula 6. By also allowing for spatial variation but no time dependence of the background seismicity rate, and estimating it within spatial Voronoi partitions, Nandan et al. (2017) estimated the ETAS model parameters for California based on earthquakes that occurred from 1981 to 2015. Through the formulaic determination of the branching ratio, the spatial distributions of K, a, and b lead to significant spatial heterogeneity in branching ratios across California, with values ranging from 0 to 1.2. Nandan et al. (2017) identified regions of California with $n > 1$ while other regions are characterized by $n < 1$, suggested to be associated with respectively negative and positive thermal anomalies. Chu et al. (2011), utilizing an ETAS model with a constant background rate and the formulaic approach, calculated a branching ratio below 1 for global regions with varied tectonic backgrounds. In more recent studies, Nandan, Ram, et al. (2021) and Nandan et al. (2022) developed an ETAS model with an improved parametric representation of the spatial variation of the background seismicity rate. Using the formulaic approach, they computed the branching ratio for global, California, and New Zealand, finding subcritical seismicity, that is, $n < 1$. In contrast, they show that assuming a uniform background leads to estimated n close to or equal to 1 in the three regions, which would lead to conclude that seismicity is operating at a critical point. In fact, they stress that this last conclusion is erroneous. Because the spatial heterogeneity of the background seismicity is not accounted for adequately when forcing a uniform background on the ETAS model calibration, the calibration of the ETAS model is driving the estimated n values close to 1, which is the domain of parameters for which the ETAS model produces the strongest spatial and temporal clustering. In other words, criticality is only apparent and is spuriously inferred in the ETAS model calibration as being the statistical bias needed for the mis-specified ETAS model to account for the pre-existing clustering of the underlying background seismicity.

It is useful to summarize the insights obtained from another field (financial time series) in which the effect of non-constancy of the background rate also distorts significantly the estimation of the branching ratio, thus showing that the effect is common to both the spatial and temporal domains. Wheatley et al. (2019) and Wehrli et al. (2021) documented the same effect in the time domain, that is, when imposing a constant background in the ETAS model calibration (in the field of finance as well as in mathematics, the model is called the Hawkes model) of a sequence of events with non-constant background. This question arises in the determination of whether financial markets operate or not in a critical state. The controversy started with the report of Filimonov and Sornette (2012) that the branching ratio (now only using sequences of events in the time domain) of financial time series is clearly less than 1, disqualifying criticality. Shortly after, Hardiman et al. (2013) countered with their finding that n is very close to 1, confirming criticality. Among others, the main difference between the two works is the length of the calibrated time series: Filimonov and Sornette (2012) used time series of 10–30 min durations (with typical waiting times between events of the order of seconds) in order to minimize biases from non-stationarity. In contrast, Hardiman et al. (2013) used much longer time series of up to several months, with the rationale of being better able to reveal a potential critical behavior expressing itself at large temporal scales. Both works calibrated the temporal ETAS model with a constant background. This is where the works of Wheatley et al. (2019) and Wehrli et al. (2021) illuminated the debate. Because financial time series are highly non-constant in time, with high volatility at the opening and at the closing of each day of trading, and with low volatility at lunch time, calibrating such time series extending over several days with an ETAS model constrained to have a constant background pushes spuriously the branching ratio to the critical value 1, so as to model the unaccounted for strong non-stationarity. By using a sophisticated parameterization of the non-constant background with the expectation-maximization (EM) method applied to long time series of several months as in Hardiman et al. (2013), Wheatley et al. (2019), and Wehrli et al. (2021) ruled out in favor of the conclusion of Filimonov and Sornette (2012). It is now understood that, by using short time series, the estimation of the latter authors was not susceptible to the bias induced by non-stationarity.

2.4. Effects Biasing the Branching Ratio *n*

It is important to note that the three existing methods for estimating the branching ratio have different pros and cons. In comparison with the other two methods, the mean-variance-based estimation approach significantly simplifies the estimation of the branching ratio. However, Wheatley et al. (2019) and Wehrli et al. (2021) demonstrated that this method gives strongly misleading estimations of the branching ratio when the true background rate is non-constant, similarly to the bias identified by Nandan, Ram, et al. (2021) when the background rate is spatially non-uniform, as discussed above. Additionally, the counting approach and the mean-variance-based estimation approach are only suitable for $n < 1$. While the formulaic approach is theoretically applicable in any case, its reliability is highly contingent on accurately estimating parameters b, K, and a, along with reasonable assumptions for m_0 and m_{max}. In contrast, the counting approach relies on accurate estimates of N_{tri} or N_{bkg}. The differences in the branching ratio estimations based on these methods are elaborated in Section 5.1.

In addition to the biases induced by non-stationarity and non-uniformity of the background rate, the three aforementioned estimation methods generally neglect or incorrectly address three crucial factors.

1. The boundary effect arises from the use of earthquake catalogs with limited duration and spatial extent. This can lead to a shift in estimation of n, as discussed in works by Wang et al. (2010) and Seif et al. (2017), and is exemplified in another context in the line-percolating system depicted in Figure 4 of Vanneste et al. (1991);
2. The finite-size effect stems from the inherently limited statistical size of the seismicity sample. These limitating effects typically occur alongside other effects and can result in both large variance and bias in the estimation of n, as shown in studies by Sornette & Utkin (2009) and Seif et al. (2017), and is also illustrated in the line-percolating system shown in Figure 3 of Vanneste et al. (1991); and
3. Censorship refers to the missing events that are not taken into account to calibrate the ETAS model due to being too small to be measured but which do trigger earthquakes. Recall that, in the ETAS model, all earthquakes of magnitude larger than a minimum magnitude m_0 can trigger "daughters" and thus impact the whole seismicity. This m_0 has no reason to be equal to the catalog cut-off magnitude M_{co} above which the catalog is deemed to be complete (Saichev & Sornette, 2006; Seif et al., 2017; Sornette & Werner, 2005b) and is analyzed. Indeed, in the ETAS model, m_0 is supposed to be a physical property of earthquake interactions. In contrast, M_{co} is usually conservatively set larger than the magnitude completeness M_c, which is constrained by instruments, that is, by the number and coverage of seismic stations and their sensitivity (e.g., Li et al., 2023). As larger investments in more numerous stations are made and better technology is developed, M_c is continuously pushed to smaller values, see for example, Feng et al. (2022) and Li et al. (2023). Then, it is likely that m_0 is smaller than M_{co}, which leads to the censorship bias. However, even large conservative estimates of completeness magnitude do not ensure 100% completeness (e.g., due to short-term aftershock incompleteness), which can introduce additional bias.

Recall that the assumption of a finite value for m_0 is crucial for ensuring that seismicity remains bounded in the ETAS model and in most of its variants (Sornette & Werner, 2005a). When m_0 is finite and when the Gutenberg-Richter and fertility laws are expressed consistently together with m_0 being their lowest magnitude of validity, then the branching ratio n is independent of m_0. On the other hand, if the lowest magnitude at which both Gutenberg-Richter and fertility laws apply goes to minus infinity, the branching ratio diverges (is no more defined). Indeed, if m_0 is pushed toward more and more negative values (smaller and smaller corresponding energies), the number of very small earthquakes grows without bound and the ETAS model is no more well-defined in the standard regime where the base-10 fertility exponent $a = \alpha / \ln(10)$ is smaller than the Gutenberg-Richter b-value.

The boundary effect and censorship both contribute to the omission of earthquakes that should be considered in a correct calibration of ETAS model. Due to the boundary effect, earthquakes outside the spatio-temporal domain of investigation (the so-called primary catalog) are not considered, while they can be potential "mothers" and "daughters" of earthquakes in the primary catalog. Censorship refers to the fact that all earthquakes of magnitudes between m_0 and M_{co} are also missing in the calibration of the ETAS model. But they are also potential "mothers" and "daughters" of the detected earthquakes with $m \geq M_{co}$. These effects disrupt the determination of possible triggering relationships between earthquakes, introducing biases and increasing variance (e.g., finite-size effect) in parameter estimation, particularly in branching ratio estimation (Saichev & Sornette, 2006; Seif et al., 2017; Sornette & Utkin, 2009; Sornette & Werner, 2005b).

In the presence of these effects, the branching ratios that have been estimated in the literature can be considered to be all biased, giving apparent branching ratios n_{app} that are likely different, and perhaps very different, from the true unknown branching ratio n_{true} (Figure 1). This may lead to potentially strongly misleading inferences on the nature of crustal seismicity. The boundary effects is a primary reason for obtaining estimated apparent branching ratios n_{app} less than 1, when using the counting and mean-variance-based estimation methods. In the comparison between n_{true} and n_{app} estimated using the mean-variance-based estimation approach, Hardiman and Bouchaud (2014) demonstrated in their Figure 1, using simulated data, that as n_{true} approaches 1, n_{app} starts to deviate significantly from n_{true}. This result is primarily influenced by the boundary effect in the time domain, as their

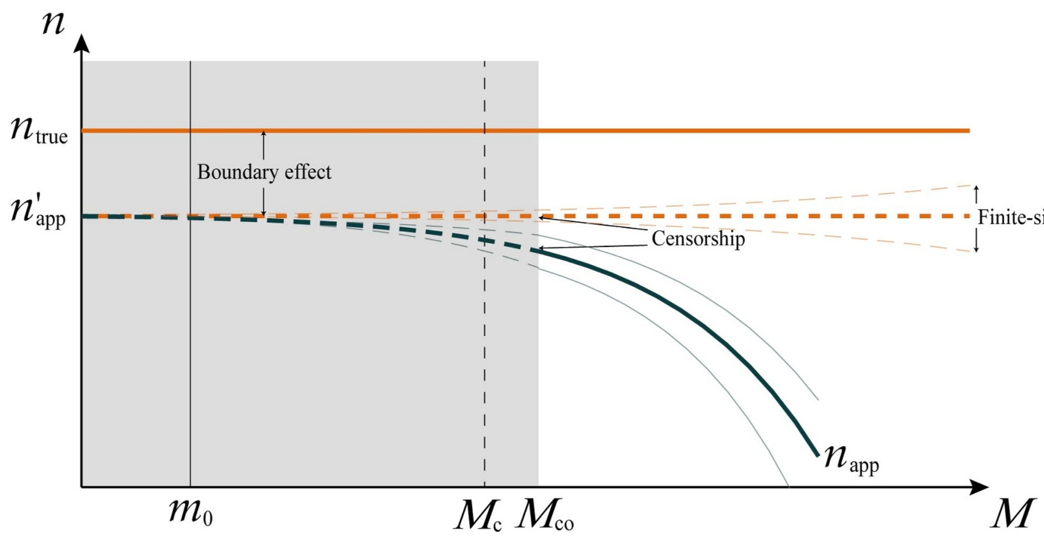

Figure 1. Schematic diagram illustrating the biases that transform the true branching ratio (n_{true}) into the apparent branching ratio (n_{app}). This diagram abstracts from common features observed in simulation experiments. In these experiments, n_{true} (orange solid line) is downwardly/upwardly biased to n'_{app} (orange dashed line) due to the boundary effect in time and/or space, with its standard deviation (orange dashed thin line) influenced by the finite-size effect. n_{app} (dark teal green dashed/solid line) is further biased downward from n'_{app} due to censorship related to the cut-off magnitude (M_{co}), which is empirically and conservatively set slightly above the completeness magnitude (M_{c}), with its standard deviation (dark teal green dashed/solid thin line) also affected by the finite-size effect. Recall that the smallest magnitude of earthquakes that can trigger other earthquakes is represented m_0, as defined in Formulas 2 and 3 respectively. In order to simplify model calibration, it is the standard practice to assume that M_{co} is equal to m_0.

simulation had no spatial component and they were modeling data over a time length of 100,000 s. The impact of censorship is also substantial. Chu et al. (2011) and Nandan, Ram, et al. (2021) obtained a branching ratio far below 1 for the global catalog, primarily because they both selected $M_{\text{co}} = 5$, leading to results likely to be significantly influenced by censorship (Saichev & Sornette, 2006; Seif et al., 2017; Sornette & Werner, 2005b), as we are going to demonstrate quantitatively below.

2.5. Existing Methods to Correct the Apparent Branching Ratio n_{app}

A few previous authors have acknowledged the impact of the boundary effect, the finite-size effect and censorship and have proposed correction procedures for branching ratio estimation. Sornette & Werner (2005b) hypothesized that earthquakes affected by censorship, due to the absence of their "mother" earthquakes, would be classified as background events. Assuming a known m_0 and a complete catalog above M_{co}, they introduced a censorship correction relating n_{app} from the counting approach to n_{true}, formulated as follows:

$$n_{\text{true}} = \begin{cases} n_{\text{app}} \left[\dfrac{10^{(b-a)(m_{\max}-m_0)} - 1}{10^{(b-a)(m_{\max}-M_{\text{co}})} - 1} \right] & a \neq b \\ n_{\text{app}} \left[\dfrac{m_{\max} - m_0}{m_{\max} - M_{\text{co}}} \right] & a = b \end{cases} \tag{13}$$

This formula is such that $n_{\text{app}} = n_{\text{true}}$ for $m_0 = M_{\text{co}}$. Seif et al. (2017) also recognized biases in branching ratio estimation due to censorship and boundary effects. They proposed a formula to derive n_{true} from n_{app} obtained from the formulaic and counting approaches in the presence of the boundary effect in the time domain, which reads:

$$n_{\text{true}} = \frac{n_{\text{app}}}{1 - \delta_{\text{TBE}}} \tag{14}$$

Here δ_{TBE}, derived analytically, represents a boundary effect correction factor in the time domain. This factor is dependent on the Omori parameters of the ETAS model and the temporal duration of the catalog. However, their analysis did not explore boundary effect in the spatial domain and failed to differentiate the distinct effects of boundary and finite-size of seismicity sample on n_{app}. Notably, finite-size effects primarily impact the standard deviation of estimations. While Seif et al. (2017) recognized the impact of censorship on branching ratio estimation, they did not propose any correction method for this issue. Moreover, their simulation experiments, exploring n_{app}'s variation with M_{co} in simulated catalogs, assumed branching ratios as high as $n = 6$ for $M_{\mathrm{co}} = 2.5$ and $n = 25$ for $M_{\mathrm{co}} = 2$, contradicting realistic physical constraints that prevent excessively high values to avoid explosive outcomes. Recall that a stationary ETAS process requires $n \le 1$.

While the corrections proposed by Sornette & Werner (2005b) and Seif et al. (2017) seem sensible, they fail to recognize the impact of a fundamental property of the ETAS model on its calibration: clustering in space and time. Specifically, Werner (2008) pointed out in his Chapter 4 that calibrating the ETAS for seismicity with $M_{\mathrm{co}} > m_0$ gives smaller changes of K (and thus of n) than predicted by Sornette & Werner (2005b). Indeed, the calibration of ETAS models is in fact controlled by the existence of spatio-temporal clustering, rather independently of the specific genealogy of which earthquake triggers what earthquake. In other words, triggered earthquakes that lose their "mothers" due to their unobservability (via boundary effects and/or censorship) may not be systematically categorized as background but may be correctly seen as part of a triggered cluster. This is due to the presence of other earthquakes in the same family tree within the cluster. To stress this fundamental property again, the genealogy within an earthquake sequence generated by the ETAS model is intrinsically stochastic since the mapping of the ETAS model to branching processes holds in the sense of an ensemble of trees where each possible filiation occurs with probabilities determined from the triggering kernels of the ETAS model. This makes the specific attribution of "motherhood" mostly irrelevant for the problem of parameter estimation. The key concept is that ETAS models account for clustering and only clustering.

3. Data-Driven Framework for Correcting Bias in Branching Ratio *n*

We now introduce a bias correction framework for the branching ratio driven by a large data set of simulated earthquake catalogs. The framework is used first to correct for censorship and finite-size effects, restoring $n_{\mathrm{app}}(M_{\mathrm{co}})$ to n'_{app}, and then to further correct for boundary and finite-size effects, bringing n'_{app} back to n_{true}. Recall that finite size effects always occur alongside other effects, resulting in both large variance and bias in the estimation of the branching ratio. Therefore, in our framework, we address this effects jointly with censorship and boundary effects.

3.1. Synthesis of Simulated Earthquake Catalogs

Employing the ETAS model defined in Section 2.1, we generate synthetic catalogs for the study regions (Table S1 in Supporting Information S1). In the present study, the spatial and temporal extent used for synthesizing the earthquake catalogs matches that of the primary catalog, but the simulated catalogs are generated for the period immediately following the last day of the primary catalog (Figures S1–S5 and Table S1 in Supporting Information S1). Therefore, the synthesized catalogs consist of three components: the earthquakes triggered by events in the primary catalog, background earthquakes occurring within the simulated window, and the aftershocks triggered by these background events. We simulate synthetic catalogs for the study regions, selecting parameters based on insights gained from subsequent calibrations performed on real catalogs. To simulate a variety of scenarios, we vary the parameters K and α of the fertility law within the ranges of 0.2085–0.9931 and 0.0084–0.4601, respectively, allowing us to model earthquake catalogs under different n_{true}. The selected ranges for K and α are based on the observed values from the $K(M_{\mathrm{co}})$ and $\alpha(M_{\mathrm{co}})$ curves across all scenarios in the study regions. Utilizing the calibrated Q, D, and IP at $M_{\mathrm{co}}^{\mathrm{best}}$ and using Formula 5 for the background rate, we generate the probability density function $\mathrm{PDF}_{\mu(x,y)}$ for background earthquakes. The number of background earthquakes is determined using the Gutenberg-Richter relationship fitted above $M_{\mathrm{co}}^{\mathrm{best}}$ to calculate the total expected number of earthquakes with magnitude greater than or equal to m_0, and then using the intended n_{true} to calculate the expected number of background earthquakes within that total. With this number as the parameter for a Poisson distribution, 100 random background earthquake counts are generated, resulting in 100 sets of background earthquake catalogs for each n_{true}. These synthesized background earthquakes trigger cascading earthquakes according to the laws of the ETAS model.

3.2. Correction of Censorship and Finite-Size Effects

The synthetic catalogs can be used to identify and quantify three types of biases affecting the branching ratio n: boundary effects in time and space, finite-size effects, and censorship. They also drive the models designed to

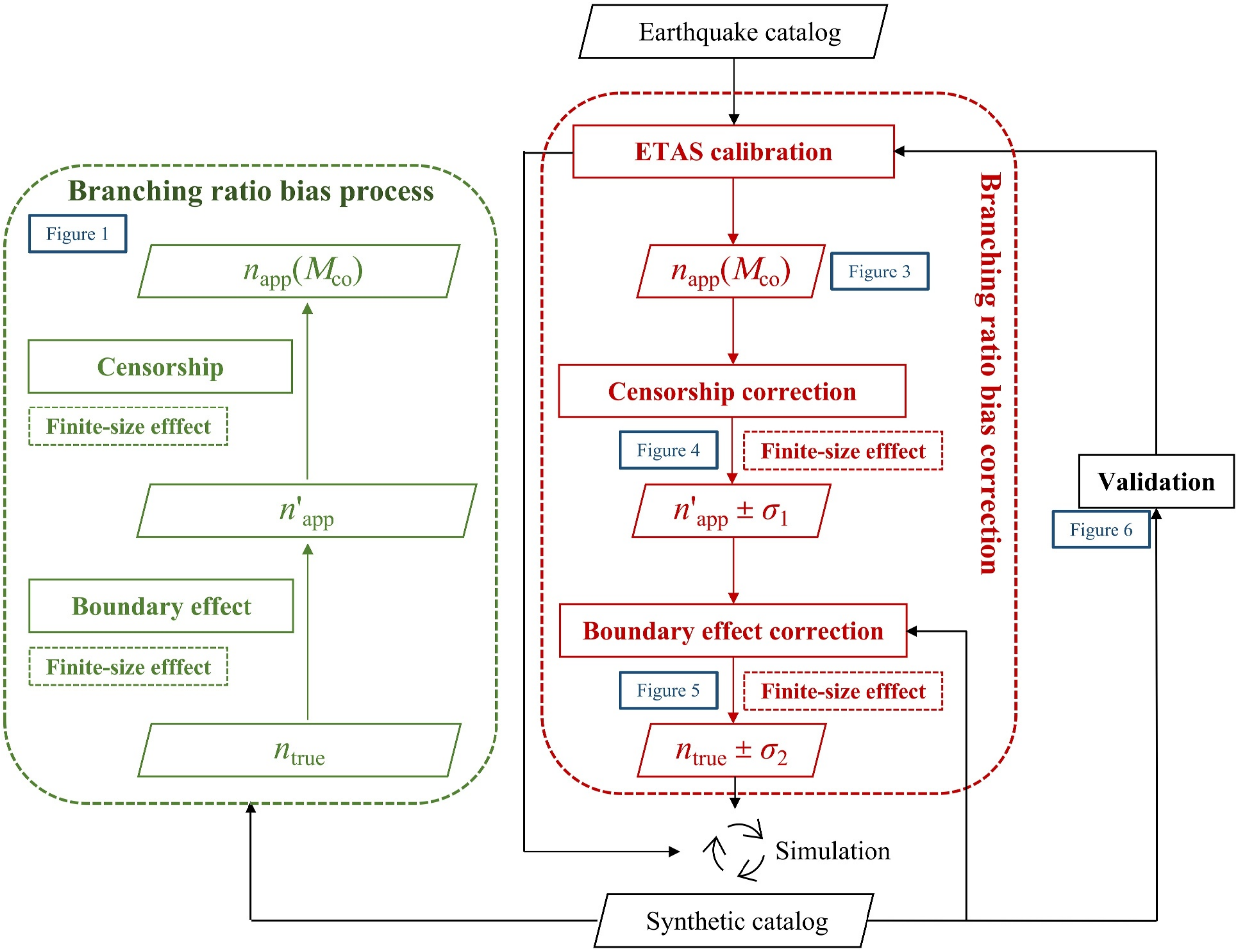

Figure 2. Flowchart of the branching ratio bias correction proposed in the present study. The green dashed box on the left illustrates by using simulation experiments how the true branching ratio n_{true} is biased due to boundary effects, finite-size effects, and censorship, progressively shifting to the prime apparent branching ratio n'_{app} and the final apparent branching ratio $n_{app}(M_{co})$ (as shown in Figure 1). The red dashed box in the center represents the process to correct the branching ratio bias, which consists of three key components: (1) robustness assessment of different branching ratio estimation methods (Figure 3), (2) correction for censorship and finite-size effects (Figure 4), and (3) correction for boundary and finite-size effects (Figure 5). Using the ETAS model calibration parameters obtained from the observed catalog for the magnitude threshold M_{co}^{best}, we generate synthetic earthquake catalogs with different true branching ratios n_{true} by varying parameters K and α. These synthetic catalogs serve three primary purposes: (a) to quantify the bias process (Figure 1), (b) to construct correction models for boundary and finite-size effects, and (c) to validate the effectiveness of the bias correction.

correct for these effects. All synthetic catalogs exhibit systematic branching ratio bias processes, which are summarized in Figure 1. Broadly, the biases affecting the branching ratio can be divided into three categories: (a) spatial and temporal boundary effects from catalog selection lead to a downward/upward bias from the theoretical true branching ratio n_{true} to the prime apparent branching ratio n'_{app}; (b) the existence of a cut-off magnitude M_{co} for calibrating models introduces a further downward bias from n'_{app} to $n_{app}(M_{co})$; (c) finite-size effects due to finite earthquake sample sizes result in statistical fluctuations of the estimated n'_{app} and n_{app} whose standard deviations are all the larger, the smaller the earthquake data sets.

The properties of the biases described in Figure 1 allow us to propose a correction method that begins with $n_{app}(M_{co})$, progresses to n'_{app}, and finally yields a refined estimate for n_{true} (Figure 2). This correction method exploits the information contained in the function $n_{app}(M_{co})$, which is much richer than the point estimate of n_{app} at M_{co}^{best}. Our correction method starts by constructing the function $n_{app}(M_{co})$ that quantifies the impact of censorship. Through observation and testing of simulated data, we propose to fit $n_{app}(M_{co})$ by the model

$$n_{app}(M_{co}) = p_1 \ 10^{p_2 M_{co}} + n'_{app} + x_1 \sigma_1 \ [N(m \geq M_{co})] \tag{15}$$

where $p_1 < 0$ and $p_2 > 0$ are two scalar parameters and the corrected intermediate (or prime apparent) branching ratio n'_{app} ($>n_{\mathrm{app}}(M_{\mathrm{co}})$) is obtained as a calibrated parameter, together with p_1 and p_2. For a given function $n_{\mathrm{app}}(M_{\mathrm{co}})$ that depends on the geometry of the auxiliary spatial domains, on the auxiliary temporal bands and on the other seismological parameters specific to the studied catalog, our procedure consists in fitting Formula 15 to the function $n_{\mathrm{app}}(M_{\mathrm{co}})$ obtained by calibrating the ETAS model and thus obtain the partially corrected n'_{app}. The model quantifies how n_{app} moves away from n'_{app} as M_{co} is increased. Inverting this relationship provides an estimate correction of $n_{\mathrm{app}}(M_{\mathrm{co}})$ into n'_{app}. Accounting for the finite-size effects, $\sigma_1[N(m \geq M_{\mathrm{co}})]$ is the standard deviation of $n_{\mathrm{app}}(M_{\mathrm{co}})$ associated with the finite number of earthquakes of magnitudes larger than M_{co}. x_1 is a random variable with zero mean and unit variance, expressing that $n_{\mathrm{app}}(M_{\mathrm{co}})$ contains a stochastic component. In the correction framework, $\sigma_1[N(m \geq M_{\mathrm{co}})]$ is defined as the boundary value of the 99% confidence interval for the parameter estimate of n'_{app} during the model fitting process.

The censorship correction model also provides information about m_0, the smallest magnitude of earthquakes capable of triggering other earthquakes defined in Formulas 2 and 3. We propose that the largest M_{co} below which the curve of n_{app} starts to flatten as M_{co} decreases corresponds to m_0, as sketched in Figure 1. The reasoning is that including smaller earthquakes in the ETAS calibration does not alter the branching ratio, indicating that we accurately account for all triggering earthquakes above this characteristic magnitude. It is determined in the present study as the minimum M_{co} at which the derivative of the censorship correction formula $n_{\mathrm{app}} = p_1 10^{p_2 M_{\mathrm{co}}} + n'_{\mathrm{app}}$ is larger than the threshold -0.01. Thus, the estimation uncertainty of m_0 is defined as the uncertainty in the M_{co} value when the derivative is -0.01, given by the formula according to the error propagation theory:

$$\sigma_{m_0} = \sigma_{M_{\mathrm{co}}}^{\mathrm{derivative}=-0.01} = \sqrt{\left(\frac{\partial M_{\mathrm{co}}}{\partial p_1}\sigma_{p_1}\right)^2 + \left(\frac{\partial M_{\mathrm{co}}}{\partial p_2}\sigma_{p_2}\right)^2 + \left(\frac{\partial M_{\mathrm{co}}}{\partial n'_{\mathrm{app}}}\sigma_1\right)^2} \tag{16}$$

where

$$\begin{cases} \dfrac{dM_{\mathrm{co}}}{dp_1} = -\dfrac{1}{p_1 \ln(10) \cdot p_2} \\ \dfrac{dM_{\mathrm{co}}}{dp_2} = -\dfrac{\log_{10}(n_{\mathrm{app}} - n'_{\mathrm{app}}) - \log_{10}(p_1)}{p_2^2} \\ \dfrac{dM_{\mathrm{co}}}{dn'_{\mathrm{app}}} = -\dfrac{1}{p_2 (n_{\mathrm{app}} - n'_{\mathrm{app}}) \ln(10)} \end{cases} \tag{17}$$

3.3. Correction of Boundary and Finite-Size Effects

The last step consists in identifying the relationship between n_{true} and n'_{app} (Figure 2). For $n_{\mathrm{app}}(M_{\mathrm{co}})$ calculated using the counting approach Formula 8, and after correcting for censorship and finite-size effects to obtain n'_{app} (the reasons for using Formula 8 to calculate $n_{\mathrm{app}}(M_{\mathrm{co}})$ are discussed in Section 5.1), it essentially represents the proportion of triggered earthquakes among the total number of earthquakes. Therefore, n'_{app} tends toward 1. If n'_{app} equals 1, it implies that all earthquakes in the catalog are considered triggered. Our simulation experiments indicate that this relationship is approximately

$$n_{\mathrm{true}} = c_2 - c_1 \cdot \log\left[\exp\left(\frac{1 - n'_{\mathrm{app}}}{c_1}\right) - 1\right] + x_2 \sigma_2 [N(m \geq m_0)] \tag{18}$$

However, when the spatiotemporal window is sufficiently large, this relationship can be approximated by a linear dependence, which leads to our other proposed correction

$$n_{\mathrm{true}} = q_1 \; n'_{\mathrm{app}} + q_2 + x_2 \; \sigma_2 \; [N(m \geq m_0)] \tag{19}$$

where c_1 and c_2, as well as q_1 and q_2 are two sets of positive scalar parameters, respectively.

Accounting for the finite-size effects, $\sigma_2[N(m \geq m_0)]$ is the standard deviation of n_{true} associated with the finite number of earthquakes of magnitudes larger than m_0. x_2 is a random variable with zero mean and unit variance, expressing that n_{true} contains a stochastic component. According to error propagation theory, we use the following formula to estimate the standard deviation of n_{true},

$$\sigma_2 \approx \begin{cases} \left| \dfrac{\mathrm{d}\left(c_2 - c_1 \cdot \log\left[\exp\left(\dfrac{1-n'_{\mathrm{app}}}{c_1}\right)-1\right]\right)}{\mathrm{d}n'_{\mathrm{app}}} \right| \cdot \sigma = \dfrac{\exp\left(\dfrac{1-n'_{\mathrm{app}}}{c_1}\right)}{\exp\left(\dfrac{1-n'_{\mathrm{app}}}{c_1}\right)-1} \cdot \sigma; \text{ for Formula(18)} \\ \left| \dfrac{\mathrm{d}\left(q_1 n'_{\mathrm{app}} + q_2\right)}{\mathrm{d}n'_{\mathrm{app}}} \right| \cdot \sigma = q_1 \cdot \sigma; \text{ for Formula(19)} \end{cases} \tag{20}$$

where,

$$\sigma = \begin{cases} \sqrt{\sigma_{c_1}^2 + \sigma_{c_2}^2 + \sigma_1^2}; \text{ for Formula(18)} \\ \sqrt{\sigma_{q_1}^2 + \sigma_{q_2}^2 + \sigma_1^2}; \text{ for Formula(19)} \end{cases} \tag{21}$$

Our method thus establishes the relationship between $n_{\mathrm{app}}(M_{\mathrm{co}})$ and n_{true}. We will apply this bias correction framework to four observed catalogs. Upon obtaining n_{true} from the estimated $n_{\mathrm{app}}(M_{\mathrm{co}})$, as a consistency check, we generate again synthetic catalogs with this n_{true} estimate and reapply the bias correction framework to these catalogs to validate that our correction from $n_{\mathrm{app}}(M_{\mathrm{co}})$ for the real catalog to n_{true} is accurate (Figure 2), as elaborated in Section 5.4.

4. Data

The earthquake catalogs utilized in the present study encompass four distinct regions (Figure S1 in Supporting Information S1): California (U.S. Geological Survey, 2017), New Zealand (GNS Science, 2022), the CSES in Sichuan and Yunnan provinces (Wu, 2022; Wu & Li, 2021a, 2021b), and Noto peninsula of Japan (Peng et al., 2024). The California catalog is sourced from the Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat). For the catalog in New Zealand, the data are obtained from the GeoNet Earthquake Catalog of New Zealand. The data for the China Seismic Experimental Site are acquired from the China Earthquake Networks Center (CENC). The relocated earthquake catalog for the Noto Peninsula since 2018 can be found in the appendix of Peng et al. (2024). The earthquake catalogs used in the present study for California, New Zealand, and CSES involve a mixture of different magnitude types. Specifically, the California catalog includes local magnitude (M_{L}), body-wave magnitude (M_{b}), moment magnitude (M_{W}), and duration magnitude (M_{d}); the New Zealand catalog reports M_{L}, M_{b}, and M_{W}; and the CSES catalog includes M_{L} and surface magnitude (M_{S}). Therefore, any research using these three data sources is likely to encounter the issue of mixing different magnitude types. In contrast, the earthquake catalog for the Noto Peninsula in Japan contains only the Japan Meteorological Agency magnitude (M_{JMA}), which is computed using body-waves with a period of less than 30 s. The magnitude bin for earthquakes is set at 0.1.

The data set spans from 1 January 1970 to 27 October 2023, for California, 1 January 1970 to 12 December 2023, for New Zealand, 1 January 1970 to 23 August 2023, for CSES, and 1 January 2018 to 19 February 2024, for Noto of Japan. For each region, earthquakes with depths down to 100 km and with $M \geq 0$ are included, as shown in Figures S2 to S5 in Supporting Information S1. Figures S2 to S5 in Supporting Information S1 also present the statistical characteristics of these earthquake catalogs. The completeness magnitude (M_{c}) is calculated using a 1-year moving step with a 2-year window. The method for quantifying power-law behavior of the complementary cumulative frequency-magnitude distribution, as proposed by Clauset et al. (2009), is employed for this calculation. Additionally, the variation of the b-value with M_{co} is explored, determined through the maximum likelihood method proposed by Aki (1965). The $\pm 1\sigma$ standard deviation of the b-value is obtained through bootstrapping (Efron, 1979). Based on the results, the primary catalog supplied to the ETAS model starts in 1985 for California, in 1990 for New Zealand, in 2000 for the CSES, and in 2020 for Noto of Japan each with an auxiliary time band of 15, 20, 30, and 2 years, respectively (Table S1; Figures S2 to S5 in Supporting

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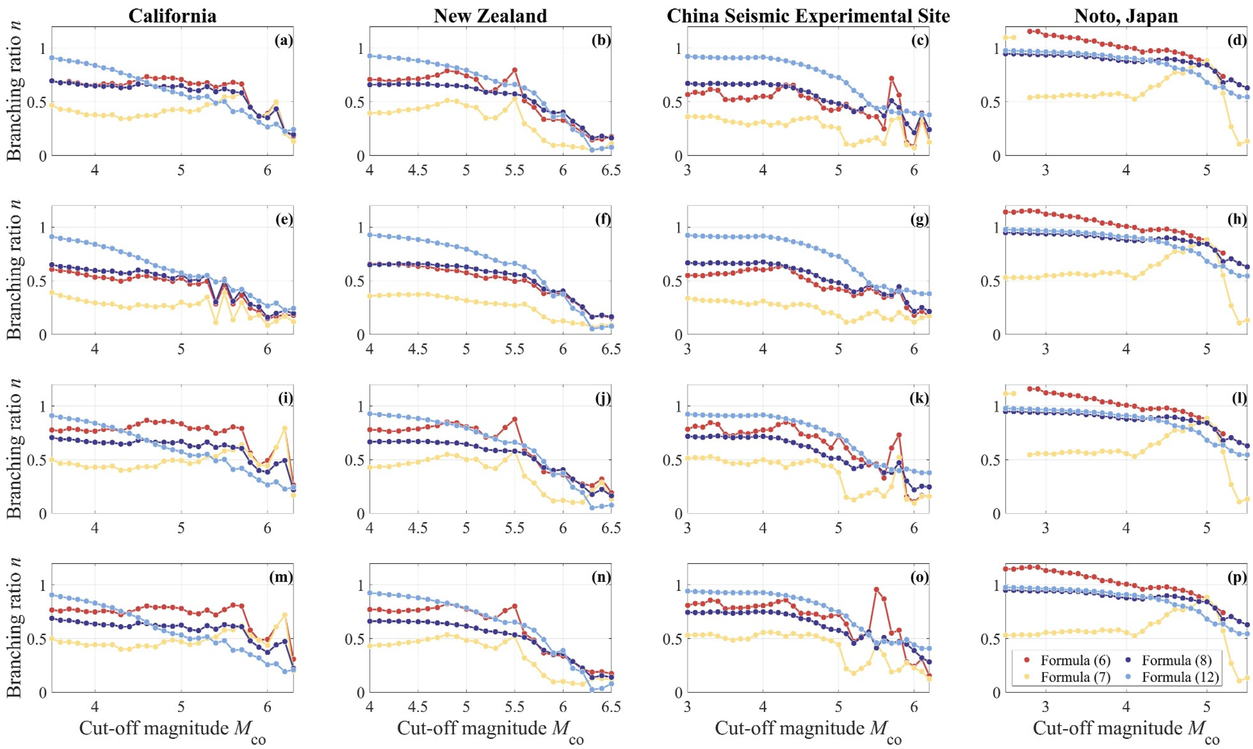

Figure 3. Estimation of the apparent branching ratio n_{app} using multiple methods, shown as a function of the cut-off magnitude M_{co} for California, New Zealand, CSES, and Noto of Japan (arranged from left to right). The panels, ordered from top to bottom, represent: (1) The primary catalog with an auxiliary spatio-temporal band; (2) The primary catalog with an auxiliary spatial band only; (3) The primary catalog with an auxiliary temporal band only; (4) The primary catalog combined with the catalog from the auxiliary region, with an auxiliary temporal band.

Information S1). The primary area for each region is outlined, accompanied by an auxiliary band extending approximately 100 km outward (Table S1; Figures S2 to S5 in Supporting Information S1). The optimal $M_{\mathrm{co}}^{\mathrm{best}}$ values for California, New Zealand, CSES, and the Noto Peninsula in Japan were chosen as 3.5, 4.0, 3.0, and 2.5, respectively. Panels (b) to (g) in Figures S2 to S5 in Supporting Information S1 demonstrate that the selection of these $M_{\mathrm{co}}^{\mathrm{best}}$ values effectively minimizes the issues of overall and short-term aftershock incompleteness while also maximizing the retention of available data.

5. Results

Our correction framework (Figure 2) systematically calculates $n_{\mathrm{app}}(M_{\mathrm{co}})$, the apparent branching ratio under a cut-off magnitude M_{co}, via the ETAS model to address biases from the true branching ratio. As introduced in Section 3, we apply censorship correction using Formula 15 and boundary effect correction using Formula 18 or 19 to ultimately obtain the true branching ratio. The uncertainty introduced by finite-size effects is quantified through error propagation theory using Formula 20. In the following sections, we first estimate $n_{\mathrm{app}}(M_{\mathrm{co}})$, the apparent branching ratio under a given cut-off magnitude M_{co}, then correct for censorship and boundary-induced biases, and finally validate the framework's efficacy through controlled simulation experiments.

5.1. Estimation of the Apparent Branching Ratio $n_{\mathrm{app}}(M_{\mathrm{co}})$

Figure 3 shows how n_{app} varies as a function of M_{co} using different estimation methods. It also tests the consistency of various n-estimation methods under four distinct scenarios (ordered from top to bottom):

- Scenario 1 involves the primary catalog with an auxiliary spatio-temporal band;
- Scenario 2 pairs the primary catalog with an auxiliary spatial band only;
- Scenario 3 uses the primary catalog with an auxiliary temporal band only; and
- Scenario 4 has a larger primary catalog, which includes both the primary catalog from Scenarios 1 to 3 and the catalog within the auxiliary spatial band, all treated as part of its primary catalog. In other words, Scenario 4 no

longer has an auxiliary spatial band, as that portion is now incorporated into its primary catalog. However, it still retains an auxiliary temporal band. The purpose of establishing Scenario 4 as a control is to investigate how different definitions of the primary catalog region influence the results.

The examined estimation methods include the formulaic approach utilizing Formulas 6 and 7, the counting approach based on Formula 8, and the mean-variance-based estimation approach in Formula 12.

In Figure 3, the estimations of the branching ratio based on the mean-variance-based estimation approach, as per Formula 12, should not be trusted when dealing with spatially non-uniform background rates in real world catalogs, as noted by Wheatley et al. (2019) and Wehrli et al. (2021). Conversely, the formulaic approach Formula 7 results in the smallest n estimates for small M_{co}. Both methods exhibit significant deviations from the estimates produced by the formulaic approach Formula 6 and the counting approach Formula 8. The latter two methods give estimates of comparable sizes and display trends in the variation of n_{app} with M_{co} that are similar to those observed in the synthetic experiments illustrated in Figure 1. The estimates from the counting approach, governed by Formula 8, exhibit a more stable curve with clearer trends compared with that of the formulaic approach Formula 6, primarily due to the reliance of the latter on the need for accurately estimated parameters b, K, and a, along with reasonable assumptions for m_0 and m_{max}. The accuracy of the counting approach hinges on precise estimates of N_{tri} or N_{bkg}.

It is useful to compare with the results of Nandan, Ram, et al. (2021), who estimated the branching ratios for California and New Zealand to be 0.79 and 0.61, respectively, using Formula 6. They used the primary catalog for both regions covering the period from 1981 to 2020 (approximately 40 years), with M_{co} set at 3.0 and 4.0 for each region, respectively. Their set-up corresponds to the Scenario 3 shown as the third row of panels in Figure 3. Scenario 3 only takes into account an auxiliary temporal band in the calibration of the ETAS model. The spatial extent of the primary catalog for California in the present study is exactly the same as that of Nandan, Ram, et al. (2021), and the time range is also similar (Table S1 in Supporting Information S1). Therefore, the branching ratio ($n = 0.79$) estimated using Formula 6 in Scenario 3 of Figure 2 in the present study, with $M_{co} = 3.5$, is basically consistent with the results of Nandan, Ram, et al. (2021). However, Figure 2 also shows that using different auxiliary catalogs can lead to significant differences. For example, considering only an auxiliary spatial band results in an estimated branching ratio of about 0.6; Scenario 1, which is considered a priori to be the most complete and involves an auxiliary spatio-temporal band, yields an estimated branching ratio of about 0.7. For New Zealand, there are significant differences between the spatio-temporal range of the primary catalog used in the present study and that of Nandan, Ram, et al. (2021). As a result, the branching ratio estimated using Formula 6 in Scenario 3 of Figure 2 for $M_{co} = 4.0$ is about 0.79, whereas the estimate by Nandan, Ram, et al. (2021) is 0.61. In the following, we will revisit seismicity criticality in the light casted by our understanding of the three biases and how their corrections will impact the results.

5.2. From $n_{app}(M_{co})$ to n'_{app}

The censorship correction proposed in the present study relies on the information provided by the dependence of the apparent estimated branching ratio (n_{app}), estimated using Formula 8, as a function of different cut-off magnitudes ($M_{co} \geq M_{co}^{best}$) as depicted in Figure 3. We use Formula 15 that has been motivated and validated on synthetic catalogs as explained in Section 3. Formula 15 describes the variation of n_{app} as a function of M_{co}, and has three adjustable parameters $\{p_1, p_2, n'_{app}\}$. For each earthquake catalog, we calibrate the ETAS model for different value of M_{co} and obtain the function $n_{app}(M_{co})$ that is specific to the geometry of the auxiliary spatial domains, to the temporal band and to the other seismological parameters specific to the studied catalog. We then fit Formula 15 via a standard nonlinear least-square fitting procedure to the function $n_{app}(M_{co})$ determined from the ETAS calibration to obtain the partially corrected branching ratio n'_{app}. In the fitting procedure of Formula 15, we assign 10 times more weight to the 20 values of M_{co} closest to M_{co}^{best} as they correspond to catalogs with larger numbers of earthquakes.

Figure 4 shows the fits of $n_{app}(M_{co})$ by Formula 15 for the four regions of California, New Zealand, CSES, and Noto of Japan for the four scenarios and for the calculation method given by Formula 8 for the branching ratio. Table 1 gives the fitted model parameter values for $n_{app}(M_{co})$ determined by Formula 8. The results show that the partially corrected branching ratios for California range from approximately 0.65 to 0.68, with estimated standard deviations between 0.01 and 0.09. The results for New Zealand are around 0.70, with an uncertainty of 0.03. For

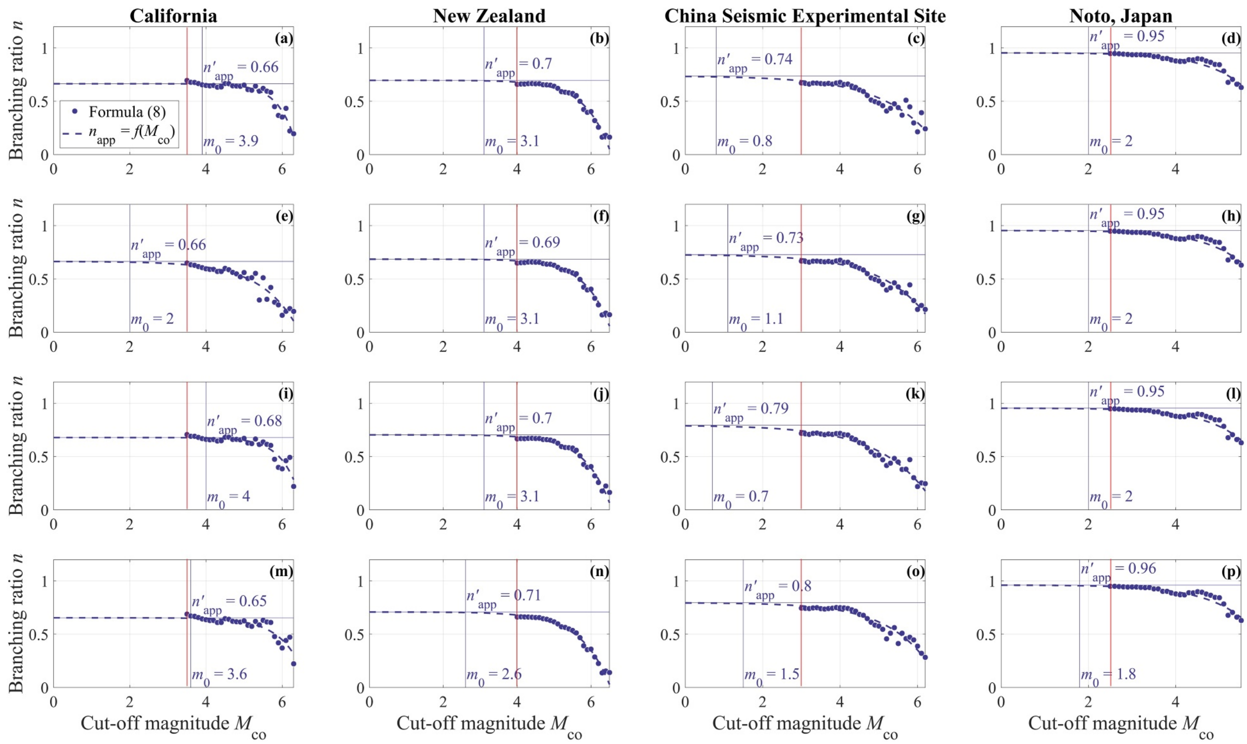

Figure 4. Correction of censorship effects using the model given by Formula 15 from which the partial corrected branching ratio n'_{app} is obtained from $n_{app}(M_{co})$ for California (left column), New Zealand (left middle column), CSES (right middle column), and Noto of Japan (right column). The four panel rows correspond to the same four scenarios as shown in Figure 3. The value of M_{co} at which the model's curve begins to level off (defined as when the derivative becomes smaller than −0.01 (larger than 0.01 in absolute value); Figure S6 in Supporting Information S1), is identified as m_0 as explained in Section 5.2. Recall that m_0 is the smallest magnitude of earthquakes capable of triggering other earthquakes. As in Figures 2 and 3, the dark blue data points are obtained with Formula 8. The red vertical line indicates M_{co}^{best}.

the CSES results, the branching ratios range from 0.73 to 0.80, with estimated standard deviations between 0.05 and 0.09. The partially corrected branching ratio for the Noto Peninsula in Japan is closest to 1, at approximately 0.95, with an uncertainty of 0.02. We hypothesize that this stability may be due to a compensatory effect at the spatiotemporal boundaries. Specifically, the number of events triggered by those outside the spatiotemporal boundaries and incorrectly identified as background events is approximately equal to the number of events that were not considered but triggered by those inside the spatiotemporal boundaries because the triggered events were located outside these boundaries. Therefore, this estimation method, which relies on the proportion of triggered earthquake counts, appears to be largely unaffected by variations in the auxiliary spatio-temporal band. Considering the need to synthesize a large number of simulated earthquake catalogs and calculate n'_{app} for each catalog in the next step, Formula 8 simply requires calculating the proportion of triggered earthquakes in the simulated catalogs, which is much simpler. Therefore, the present study will proceed with corrections using n'_{app} from Formula 8 for further analysis.

The censorship correction model also provides information about m_0, the smallest magnitude of earthquakes capable of triggering other earthquakes defined in Formulas 2 and 3. It is determined in the present study as the minimum M_{co} at which the derivative of the censorship correction formula $n_{app} = p_1 10^{p_2 M_{co}} + n'_{app}$ is larger than the threshold −0.01 (Figure S6 in Supporting Information S1). The estimated uncertainties are calculated using Formulas 16 and 17. Figure 4 and Table 1 present the estimated m_0 and its uncertainty. The results show that the estimated m_0 for California is approximately 4, with an uncertainty of 4 (Scenario 2 has a very high uncertainty of 10.5). The CSES provides the lowest estimated m_0, around 1.5, but it also has the highest uncertainty, approximately 7. In contrast, New Zealand and Noto, Japan, have the smallest estimated uncertainties, around 3.5. However, New Zealand's estimated m_0 is about 3.3, while Noto's is approximately 2.3.

Table 1
Parameters of Formula 15 $n_{app} = p_1 10^{p_2 M_{co}} + n'_{app}$ Fitted to the Function $n_{app}(M_{co})$ Determined From the ETAS Calibrations With Formula 8 of the Four Regions (Columns) for the Four Scenarios (Rows)

	California				New Zealand			
Scenario 1	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$
	4.1 ± 2.9	$-(5.08 \pm 19.79) \times 10^{-7}$	0.95 ± 0.2763	0.66 ± 0.0172	3.4 ± 3.6	$-(4.61 \pm 10.06) \times 10^{-5}$	0.64 ± 0.1518	0.70 ± 0.0317
Scenario 2	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$
	2.3 ± 10.5	$-(8.79 \pm 34.60) \times 10^{-4}$	0.44 ± 0.2792	0.66 ± 0.0917	3.4 ± 3.4	$-(4.46 \pm 9.11) \times 10^{-5}$	0.64 ± 0.1418	0.69 ± 0.0288
Scenario 3	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$
	4.1 ± 3.8	$-(8.15 \pm 41.40) \times 10^{-7}$	0.90 ± 0.3608	0.68 ± 0.0201	3.3 ± 4.1	$-(5.66 \pm 12.80) \times 10^{-5}$	0.62 ± 0.1570	0.70 ± 0.0338
Scenario 4	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$
	3.8 ± 4.5	$-(1.19 \pm 5.64) \times 10^{-5}$	0.71 ± 0.3390	0.65 ± 0.0271	2.9 ± 4.2	$-(1.84 \pm 3.24) \times 10^{-4}$	0.55 ± 0.1219	0.71 ± 0.0357
	CSES				Noto, Japan			
Scenario 1	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$
	1.3 ± 9.0	$-(4.84 \pm 11.43) \times 10^{-3}$	0.33 ± 0.1665	0.74 ± 0.0807	2.3 ± 3.2	$-(6.20 \pm 12.50) \times 10^{-4}$	0.49 ± 0.1654	0.95 ± 0.0228
Scenario 2	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$
	1.5 ± 6.6	$-(3.44 \pm 6.44) \times 10^{-3}$	0.36 ± 0.1333	0.73 ± 0.0596	2.3 ± 3.2	$-(6.20 \pm 12.50) \times 10^{-4}$	0.49 ± 0.1654	0.95 ± 0.0228
Scenario 3	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$
	1.1 ± 8.3	$-(5.23 \pm 9.84) \times 10^{-3}$	0.33 ± 0.1330	0.79 ± 0.0745	2.3 ± 3.2	$-(6.30 \pm 12.68) \times 10^{-4}$	0.49 ± 0.1650	0.95 ± 0.0229
Scenario 4	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$	$m_0 \pm \sigma_{m_0}$	$p_1 \pm \sigma_{p_1}$	$p_2 \pm \sigma_{p_2}$	$n'_{app} \pm \sigma_1$
	1.9 ± 5.6	$-(2.01 \pm 4.03) \times 10^{-3}$	0.39 ± 0.1438	0.80 ± 0.0513	2.2 ± 3.4	$-(9.08 \pm 18.51) \times 10^{-4}$	0.46 ± 0.1667	0.96 ± 0.0264

Note. Recall that M_{co} and m_0 are respectively the cut-off magnitude and the smallest magnitude of earthquakes capable of triggering other earthquakes.

5.3. From n'_{app} to n_{true}

As discussed in the introduction, the boundary effect is related to the temporal and spatial extent of the study region, while the finite-size effect results from the finite size of the seismicity sample. We propose to correct n'_{app} in order to recover n_{true} by synthesizing catalogs with the same spatiotemporal range. Specifically, we synthesize earthquake catalogs using calibrated ETAS model parameter sets $\{Q, D, c, \omega, \tau, d, \gamma, \rho\}$ at M_{co}^{best} (Table S2 in Supporting Information S1), combined with the aforementioned estimation of m_0. Since the triggering relationships among earthquakes in the simulated catalogs are known, it is straightforward to calculate n'_{app} by using Formula 8. Figure 5 shows the dependence of n'_{app} and its standard deviation as a function of n_{true} quantifying the impact of boundary effects in time and space, with standard deviation influenced by the finite-size effects, for California, New Zealand, CSES, and Noto of Japan. The mean and standard deviation of n'_{app} are calculated from 100 simulated earthquake catalogs for each n_{true}, for the four study regions under different scenarios. In Section 3, we have established through numerical simulations the relationship between n_{true} and n'_{app} for $n_{true} < 1$.

The solid lines in Figure 5 are the best fits with models in Section 3.3 to $n'_{app}(n_{true})$ for different scenarios in each study region. In the case of the Noto in Japan, we used Formula 18 for fitting. For the other three regions with larger spatiotemporal windows, we applied the linear Formula 19. Then, we selected the corresponding error estimation formulas mentioned in Section 3.3 to provide uncertainty estimates for n_{true}. Table 2 lists the values of the parameters of Formulas 18 and 19 and the corresponding final estimates of n_{true}. Accounting for boundary effects and finite-size effects, the initial estimated n_{app} needs to be corrected upwards by approximately 0.04–0.06 to obtain n_{true}. However, for Noto in Japan, a downward correction of approximately 0.1 is applied to obtain n_{true}. The upward bias of the branching ratio in Noto, from n_{true} to n'_{app} (thus needing a downward correction), is primarily due to the smaller spatiotemporal windows in the earthquake catalog. These smaller windows likely capture a significant portion of seismicity clusters, resulting in an increased proportion of triggered events in the catalog. The final n_{true} estimates are 0.71–0.74 with estimated standard deviations between 0.02 and 0.08 for California, 0.75 to 0.77 with estimated standard deviations between 0.02 and 0.03 for New Zealand, 0.79 to 0.84 with estimated standard deviations between 0.03 and 0.06 for CSES, and 0.83 to 0.85 with estimated standard

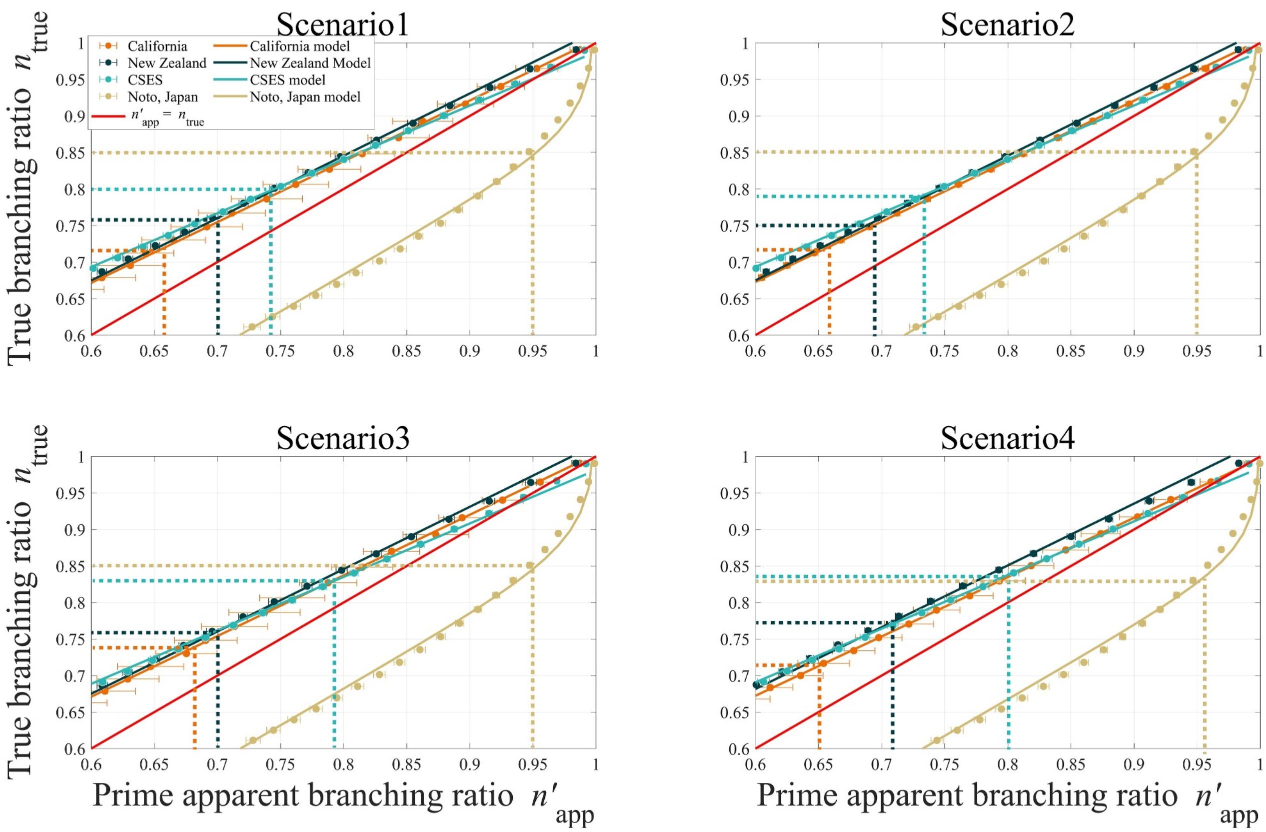

Figure 5. Dependence of n'_{app} and its standard deviation as a function of n_{true} (in reverse axis) showing the quantitative impact of boundary effects in time and space, with standard deviation influenced by the finite-size effect, for California, New Zealand, CSES and Noto of Japan. For each value of n_{true}, 100 sets of catalogs have been simulated, mirroring the time and space ranges of the primary catalog in the four study regions. The mean and variance of n'_{app} over these 100 sets of simulated catalogs are calculated using the counting approach given by Formula 8. The observed dependence suggests the correction model given by Section 3.3 which allows us to extract n_{true} from n'_{app}. The dashed lines in the figure represent the corrected n_{true} values corresponding to n'_{app} from Table 2.

deviations between 0.03 and 0.04 for Noto in Japan. The seismicity within these regions is closer to criticality than inferred without corrections (except for Noto in Japan), yet still maintains a distance of several standard deviations from the critical value $n_c = 1$, making criticality very unlikely.

Compared to the estimates by Nandan, Ram, et al. (2021) for the global ($n = 0.45$), California ($n = 0.79$), and New Zealand ($n = 0.61$), our results feature a smaller standard deviation across different scenarios and regions. The results of Nandan, Ram, et al. (2021) for the global catalog were significantly influenced by censorship due to the setting of $M_{co} = 5$. Meanwhile, their estimates for California and New Zealand did not account for boundary effects in time and space and for the finite-size effects, thus introducing some bias and uncertainty in their point estimates for these two regions. Our current findings imply that about 70%–85% of observed seismicity is triggered by preceding earthquakes, with the remaining 15%–30% being background seismicity driven by the forces of plate tectonics.

5.4. Validation of the Estimation of n_{true}

After recovering n_{true} from $n_{app}(M_{co})$, we conduct a consistency check by generating synthetic catalogs using the n_{true} estimate. We utilize m_0 estimates from censorship correction in Table 1, ETAS model calibration parameters from Table S2 in Supporting Information S1, and n_{true} in Table 2 to synthesize 100 sets of simulated earthquake catalogs for each scenario in every study region. We then reapply the bias correction framework to these catalogs (Figure 6). This process serves to validate the accuracy of our correction from $n_{app}(M_{co})$ to n_{true} for the actual catalog, ensuring the reliability of our methods.

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Table 2
Parameters for the Model Shown in Section 5.3, Utilized to Correct Boundary Effects in Time and Space, Where Standard Deviation Is Impacted by the Finite-Size Effect

	California				New Zealand			
Scenario 1	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$
	0.66 ± 0.0172	0.83 ± 0.0034	0.17 ± 0.0020	0.72 ± 0.0146	0.70 ± 0.0317	0.85 ± 0.0116	0.16 ± 0.0033	0.76 ± 0.0288
Scenario 2	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$
	0.66 ± 0.0917	0.83 ± 0.0021	0.18 ± 0.0012	0.72 ± 0.0761	0.69 ± 0.0288	0.85 ± 0.0055	0.16 ± 0.0031	0.75 ± 0.0251
Scenario 3	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$
	0.68 ± 0.0201	0.83 ± 0.0033	0.17 ± 0.0019	0.74 ± 0.0170	0.70 ± 0.0338	0.85 ± 0.0058	0.16 ± 0.0033	0.76 ± 0.0293
Scenario 4	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$
	0.65 ± 0.0271	0.81 ± 0.0015	0.19 ± 0.0009	0.71 ± 0.0220	0.71 ± 0.0357	0.85 ± 0.0073	0.17 ± 0.0041	0.77 ± 0.0312
	CSES				Noto, Japan			
Scenario 1	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$	$n'_{\mathrm{app}} \pm \sigma_1$	$c_1 \pm \sigma_{c_1}$	$c_2 \pm \sigma_{c_2}$	$n_{\mathrm{true}} \pm \sigma$
	0.74 ± 0.0807	0.74 ± 0.0060	0.25 ± 0.0036	0.80 ± 0.0599	0.95 ± 0.0228	0.04 ± 0.0047	0.88 ± 0.0063	0.85 ± 0.0338
Scenario 2	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$	$n'_{\mathrm{app}} \pm \sigma_1$	$c_1 \pm \sigma_{c_1}$	$c_2 \pm \sigma_{c_2}$	$n_{\mathrm{true}} \pm \sigma$
	0.73 ± 0.0596	0.74 ± 0.0059	0.25 ± 0.0035	0.79 ± 0.0444	0.95 ± 0.0228	0.04 ± 0.0047	0.88 ± 0.0063	0.85 ± 0.0338
Scenario 3	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$	$n'_{\mathrm{app}} \pm \sigma_1$	$c_1 \pm \sigma_{c_1}$	$c_2 \pm \sigma_{c_2}$	$n_{\mathrm{true}} \pm \sigma$
	0.79 ± 0.0745	0.73 ± 0.0072	0.25 ± 0.0043	0.83 ± 0.0547	0.95 ± 0.0229	0.04 ± 0.0047	0.88 ± 0.0063	0.85 ± 0.0339
Scenario 4	$n'_{\mathrm{app}} \pm \sigma_1$	$q_1 \pm \sigma_{q_1}$	$q_2 \pm \sigma_{q_2}$	$n_{\mathrm{true}} \pm \sigma$	$n'_{\mathrm{app}} \pm \sigma_1$	$c_1 \pm \sigma_{c_1}$	$c_2 \pm \sigma_{c_2}$	$n_{\mathrm{true}} \pm \sigma$
	0.80 ± 0.0513	0.73 ± 0.0067	0.25 ± 0.0040	0.84 ± 0.0379	0.96 ± 0.0264	0.04 ± 0.0047	0.87 ± 0.0076	0.83 ± 0.0441

Note. In this table, n'_{app} is derived by fitting the n_{app} values estimated using Formula 8, as shown in Table 1.

Since our primary goal here is to validate the accuracy of our correction method, we did not use the ETAS model to obtain $n_{\mathrm{app}}(M_{\mathrm{co}})$ for the 1,600 sets of earthquake catalogs (4 study regions multiplied by 4 scenarios multiplied by 100 catalogs). Given that we have a clear understanding of the branching structure of the simulated catalogs, we introduce a model here to simulate the ETAS estimation process for the branching ratio. In this model, the disappearance of a triggered event's "mother" due to boundary effects or censorship does not hinder the model's ability to establish a triggering relationship. Provided there is at least one "relative" within its direct paternal lineage, a connection between the triggered event and its lineage "relative" remains identifiable. This capability simulates the sensitivity of the ETAS model to clusters of seismicity. Remarkably, despite challenges posed by boundary, finite-size and censorship effects, it maintains a high degree of accuracy in identifying and classifying triggered earthquakes. Considering this model and based on Formula 8, we thus obtain $n_{\mathrm{app}}(M_{\mathrm{co}})$ for the simulated catalogs, as shown in Figure 6 by green solid points with error bar. Applying our branching ratio bias correction method to $n_{\mathrm{app}}(M_{\mathrm{co}})$, it is evident that our approach yields accurate estimates of m_0 and n_{true} (Table 3).

6. Discussion and Conclusions

We have proposed a new method to correct the three major identified biases by utilizing the functional dependence of estimated ETAS parameters on the artificial increase of the cut-off magnitude M_{co}. Our findings reveal that by accurately considering not just the spatial variation of the background rate but also censorship, temporal and spatial boundary effects, and the finite-size effects of the earthquake catalog, we obtain a branching ratio of seismicity that is closer to criticality than inferred without corrections, yet still maintains a certain distance from it, with values ranging from 0.70 to 0.85. The branching ratio n is observed to be different for tectonic regimes: it is largest in Noto of Japan and CSES, intermediate in New Zealand, and lowest in California.

The minimum magnitude m_0 for earthquake triggering capability, an intrinsic ingredient of the ETAS model, demonstrate significant variability for different tectonic regimes, with California displaying the largest m_0, followed by New Zealand and Noto of Japan, and CSES presenting the lowest. However, the estimation of m_0 exhibits significant uncertainty, indicating that the current observational data still cannot effectively constrain m_0. Additional processes may also influence the results, such as the modeling of spatial triggering using isotropic

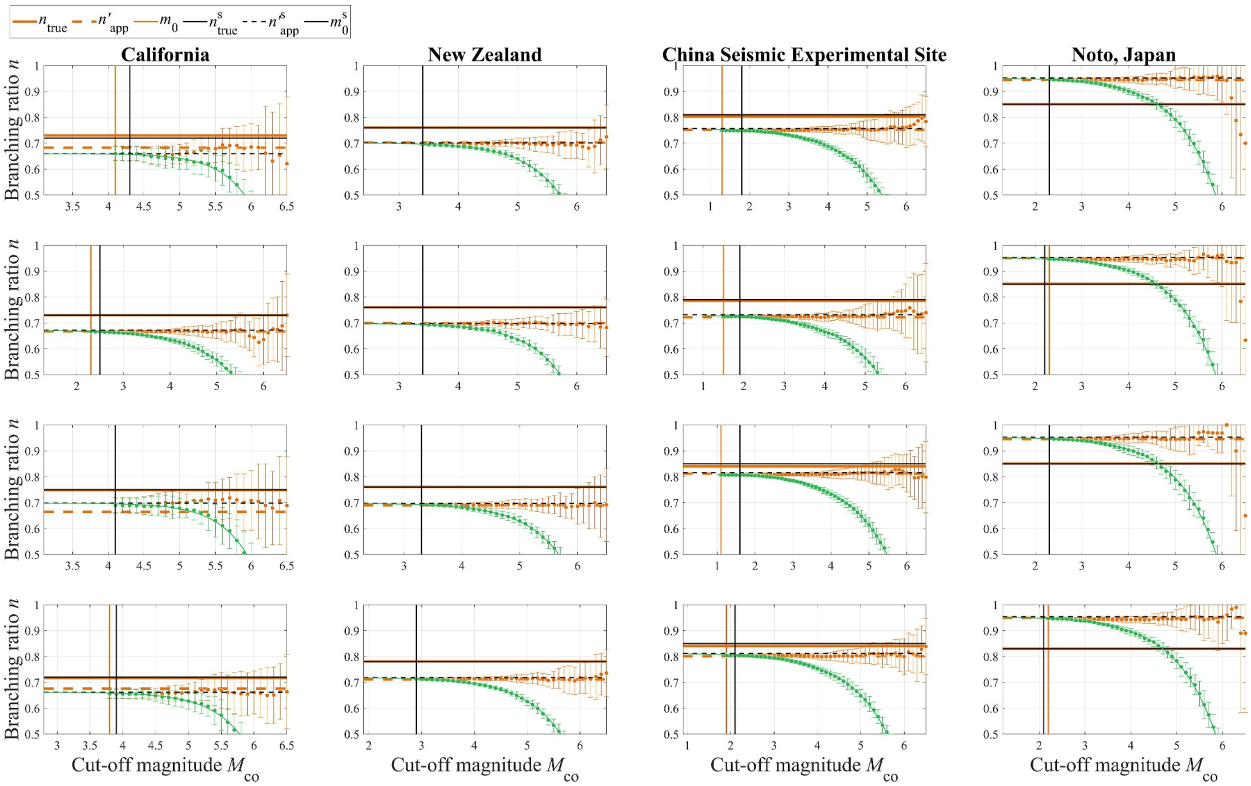

Figure 6. Validation of our bias correction method. The orange solid horizontal and vertical lines represent n_{true} and m_0 obtained from Tables 1 and 2, which are used as parameters for synthesizing 100 sets of simulated earthquake catalogs. By calculating the proportion of triggered seismicity within the spatiotemporal windows of the study regions, we derive the biased n'_{app} (orange dashed horizontal line) affected by boundary and finite-size effects. Here, n'_{app} is calculated using $M_{co} = m_0$. We then vary M_{co} to obtain the $n'_{app}(M_{co})$ (green solid points) impacted by censorship. This censorship is simulated by the model, where triggered earthquakes are correctly identified due to the presence of seismicity clusters, despite potential loss of their "mother" earthquakes, as long as other events in their triggering lineage exist. The orange circles with error bars illustrate scenarios without misclassification of triggered earthquakes as background events due to shifts in M_{co}, demonstrating that the true branching ratio (even when affected by boundary effects) is not influenced by the magnitude scale. The black dashed horizontal line, black solid horizontal line, and black solid vertical line represent n'^{s}_{app}, n^{s}_{app}, and m^{s}_{0}, respectively. They are the corrected estimates obtained from the synthetic catalogs, which were generated using n_{true}, n'_{app}, and m_0 as their true values.

kernels, despite the known anisotropic nature of aftershock distributions, or the occurrence of aseismic processes. It is important to recognize that the issue with m_0 arises in several ETAS models. Furthermore, the understanding and handling of m_0 should not rely on a simplistic truncation concept, but rather be seen as a smooth process, as the likelihood of triggering earthquakes is unlikely to drop abruptly to zero for magnitudes below m_0. Incorporating this idea in calibrations of the ETAS model is left for the future. Other models have been proposed that alleviate or even remove the need for a m_0, which incorporate two branches of the Gutenberg-Richter law, like those suggested by Vere-Jones (2005), analyzed by Saichev & Sornette (2005), and more recently observed in EM stochastic reconstructions by Nandan et al. (2022).

The non-critical but still rather large values of the branching ratio n prompt a reevaluation of our understanding of the brittle fracture process within the Earth's crust. So far, reported values of n found close to $n_c = 1$ have aligned with the popular concept of self-organized criticality, suggesting that the loaded fault network is in a permanent critical state. In contrast, values significantly below $n_c = 1$ indicate that fault networks primarily evolve far from a critical point. Our estimates of n, derived from more appropriate assumptions and an improved research methodology, fall between these two extremes, neither continuously critical nor far from it. In fact, our results align with the current state of research on earthquake predictability, suggesting that earthquakes are not entirely unpredictable as they would be if the seismogenic crust was in a self-organized critical state (where any event is mechanistically indistinguishable from others, making it impossible to effectively identify precursors before an event occurs) (Geller et al., 1997; Mizrahi et al., 2024; Schorlemmer et al., 2018). The degree of earthquake

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Table 3

n_{true} and m_0 From Real Catalogs Used for Synthesizing 100 Sets of Simulated Earthquake Catalogs, and n^s_{true} and m^s_0 Recovered by the Correction Method Based on Simulated Catalog Sets

	California				New Zealand			
Scenario 1	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$
	4.1 ± 2.9	0.72 ± 0.0234	4.3 ± 0.8	0.72 ± 0.0191	3.4 ± 3.6	0.76 ± 0.0272	3.4 ± 0.2	0.76 ± 0.0038
Scenario 2	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$
	2.3 ± 10.5	0.72 ± 0.0762	2.5 ± 0.2	0.73 ± 0.0026	3.4 ± 3.4	0.75 ± 0.0246	3.4 ± 0.3	0.76 ± 0.0027
Scenario 3	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$
	4.1 ± 3.8	0.74 ± 0.0262	4.1 ± 1.1	0.75 ± 0.0213	3.3 ± 4.1	0.76 ± 0.0290	3.3 ± 0.2	0.76 ± 0.0040
Scenario 4	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$
	3.8 ± 4.5	0.71 ± 0.0275	3.9 ± 0.6	0.72 ± 0.0169	2.9 ± 4.2	0.77 ± 0.0303	2.9 ± 0.1	0.78 ± 0.0024
	CSES				Noto, Japan			
Scenario 1	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$
	1.3 ± 9.0	0.80 ± 0.0598	1.8 ± 0.1	0.81 ± 0.0028	2.3 ± 3.2	0.85 ± 0.0321	2.3 ± 0.3	0.85 ± 0.043
Scenario 2	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$
	1.5 ± 6.6	0.79 ± 0.0442	1.9 ± 0.2	0.79 ± 0.0027	2.3 ± 3.2	0.85 ± 0.0322	2.2 ± 0.3	0.85 ± 0.0044
Scenario 3	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$
	1.1 ± 8.3	0.83 ± 0.0544	1.6 ± 0.1	0.85 ± 0.0024	2.3 ± 3.2	0.85 ± 0.0322	2.3 ± 0.3	0.85 ± 0.0038
Scenario 4	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$	$m_0 \pm \sigma_{m_0}$	$n_{\text{true}} \pm \sigma_1$	$m^s_0 \pm \sigma_{m_0}$	$n^s_{\text{true}} \pm \sigma$
	1.9 ± 5.6	0.84 ± 0.0376	2.1 ± 0.2	0.85 ± 0.0032	2.2 ± 3.4	0.83 ± 0.0419	2.1 ± 0.3	0.83 ± 0.0041

predictability may be in part associated with sporadic emergence of singularities appearing through various mechanisms that could help signal the approach of catastrophic events.

Finally, in addition to the factors discussed in this study, the fidelity of the model to actual seismicity is equally important for accurately gauging criticality. For example, including the depth component in the model formulation can greatly improve model fitting (Guo et al., 2015a, 2017; Zhuang et al., 2019), which also yields a smaller branching ratio. Another factor is the rupture geometry of large earthquakes (Guo et al., 2015b, 2019, 2021; Hainzl et al., 2008). Ignoring the volume of the earthquake's rupture but regarding the focal zone as a point leads to a larger value of K and a smaller value of α (Guo et al., 2021; Zhuang et al., 2019). Nevertheless, even the estimation for these improved versions of the ETAS model also suffers from the problems discussed in this study. Additionally, the impact of different magnitude types on the ETAS model should not be overlooked. For example, dynamic triggering may be related to high-frequency energy radiation (more closely associated with energy magnitude M_e and surface wave magnitude M_S), while static triggering may be related to low-frequency energy radiation and permanent changes in the fault's stress state (more closely associated with moment magnitude M_W). We leave these issues for future research.

Data Availability Statement

The California catalog is sourced from the Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat; U.S. Geological Survey, 2017), accessible at https://earthquake.usgs.gov/data/comcat/ (last accessed: 18 December 2023). For the catalog in New Zealand, the data are obtained from the GeoNet Earthquake Catalog of New Zealand (GNS Science, 2022), accessible at https://quakesearch.geonet.org.nz/ (last accessed: 18 December 2023). We acknowledge the New Zealand GeoNet programme and its sponsors EQC, GNS Science, LINZ, NEMA and MBIE for providing data used in this study. The data for the China Seismic Experimental Site (CSES) are acquired from the China Earthquake Networks Center (CENC) through the internal link provided by the Earthquake Cataloging System at China Earthquake Administration, available at http://10.5.160.18/console/index.action (last accessed: 18 December 2023), with a Digital Object Identifier (DOI) of 10.11998/SeisDmc/SN. This data is not publicly available; it can be requested from the CENC. The catalog of earthquakes with magnitudes of 3.0 and above in the CSES region from 1970 to 2023 is provided as a Data Set S1 to this paper. The relocated earthquake catalog for the Noto Peninsula since 2018 can be found in the appendix of Peng et al. (2024).

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